

# Veterinary AI Animal Health & Diagnostic Platform

An End-to-End Multimodal Video, Vision & Clinical Decision Support System

**Platform Version** v2.4.0 (Production Core)

**Core Backend** Go 1.22+ / SQLite WAL

**AI Architecture** Dual-Stage Multimodal LLM

**Generated Date** August 2026

## 1. Executive Summary & System Overview

The **Veterinary AI Animal Health & Diagnostic Platform** is a full-stack, multimodal clinical decision support system engineered for veterinary practitioners, veterinary technicians, agricultural livestock managers, and companion animal owners. The system ingests high-resolution still photographs and temporal live video recordings (e.g. motion, involuntary spasms, chorea, gait abnormalities, respiratory distress), extracts sequential keyframes, conducts objective computer vision inspection, and synthesizes visual evidence with owner-reported clinical history using specialized Large Multimodal Models (LMMs).

Unlike conventional veterinary AI classifiers that rigidly classify dermatology lesions or force single

disease predictions, this platform operates on an **unbiased, broad-spectrum differential diagnosis framework**. It evaluates infectious (viral, bacterial, fungal, parasitic), non-infectious (mechanical obstructions, toxicological, inflammatory, metabolic, traumatic), and behavioral presentations across canine, feline, bovine, equine, caprine, ovine, and avian species.

## 2. Problem Statement & Key Industry Challenges

Veterinary triage and diagnostics face significant operational and technical hurdles:

- **The Static Image Diagnostic Blindspot:** Many life-threatening veterinary conditions cannot be diagnosed from a single static photograph. For example, *Canine Distemper Virus (CDV)* presents with rhythmic chorea/myoclonus (jaw and leg twitching), *Equine Colic* presents with dynamic visceral pain (ground pawing, flank watching, rolling), and *Rabies* presents with behavioral agitation or ataxia. These require motion analysis across temporal video frames.
- **Classifier Bias & False Positives:** Traditional image classification models are trained on narrow dermatological datasets, frequently misclassifying critical systemic emergency conditions (e.g. Parvovirus, Distemper, AHDS) as superficial skin pyoderma or forcing predetermined diseases.
- **Data Integrity & Concurrency Lockouts:** In embedded database deployments (e.g. SQLite in veterinary clinics and field laptops), continuous automated snapshot backups frequently trigger `database is locked` errors during write-heavy clinical screening workflows.
- **Auth Header & API Token Collisions:** In multi-tier architectures, forwarding user login bearer tokens into third-party AI provider headers triggers HTTP 401/403 authorization rejections.

## 3. Our Solution & Core Innovations



### Multimodal Video & Keyframe Pipeline

In-browser camera recording (MediaRecorder) and file upload (.mp4, .webm, .mov) with automated ffmpeg

temporal sampling extracting 4 evenly-spaced keyframes across timeline adhering to AI router quotas.



### Dual-Stage Multimodal AI Pipeline

Strict separation of concerns: Stage 1 (Qwen2.5-VL-72B-Instruct) performs purely objective visual/motion

extraction; Stage 2 (Qwen2.5-72B-Instruct) executes broad-spectrum veterinary internal medicine reasoning.

 **Dual-Database & Point-in-Time Backups**

Dedicated `user.db` (Bcrypt password hashing, token auth) and `animal_health.db` (records, media, screenings)

with automatic `VACUUM INTO` point-in-time snapshot rotation upon every database mutation.

 **Lock-Free SQLite Concurrency**

WAL mode with `_busy_timeout=10000`, `PRAGMA busy_timeout = 10000;;`, and serialized write pooling

(`SetMaxOpenConns(1)`) completely eliminating lock contention during simultaneous auto-backups.

4. Complete Technology Stack & Specifications

| Tier / Component      | Technologies & Libraries                                   | Functional Responsibility                                                                                                                                                                     |
|-----------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Layer        | React 18 Vite SPA Tailwind CSS Lucide Icons                | Single-page web application featuring responsive dark/light interface, live camera video capture, drag-and-drop media upload, triage visualization, and patient records management.           |
| Client APIs & Capture | MediaRecorder API Navigator.mediaDevices Fetch API         | In-browser 5–12s gait/spasm video recording, live video preview playback, audio/video stream constraints, and custom header injection ( <code>X-Auth-Token</code> , <code>X-User-ID</code> ). |
| Backend Core          | Go 1.22+ net/http go <code>lang.org/x/crypto/bcrypt</code> | High-throughput REST API server, custom middleware pipeline (CORS, Token Auth, Request Logging), and asynchronous backup event dispatchers.                                                   |
| Media Engine          | FFmpeg ffp <code>robe</code>                               | 6.0+ Video stream validation, duration probing, temporal filter-graph scaling ( <code>fps=1/2, scale=512:-1</code> ), and 4-frame                                                             |

| Tier / Component    | Technologies & Libraries                                                   | Functional Responsibility                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                                                                            | JPEG base64 extraction.                                                                                                                                                                |
| Databases & Storage | SQLite 3 (WAL)<br>mattn/go-sqlite3                                         | Two isolated databases: <code>user.db</code> (authentication, credentials, salt/hashes) and <code>animal_health.db</code> (patients, media metadata, clinical screenings, reminders).  |
| AI Inference Layer  | Qwen2.5-VL-72B-Instruct<br>Qwen2.5-72B-Instruct<br>Hugging Face Router API | Multimodal visual analysis on keyframe arrays, followed by structured veterinary clinical assessment, differential ranking, confirmatory test prescription, and triage urgency rating. |

## 5. End-to-End System Architecture & Data Pipelines

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-----+ | FRONTEND CLIENT (React 18 SPA) | | [Photo Capture / File Drag-Drop] [In-Browser Video

Recorder (MediaRecorder)] [Symptoms Form] | +-----

+-----+ | Multipart POST /api/health-check/upload

(Headers: X-Auth-Token, X-User-ID) v

+-----

-----+ | BACKEND INGESTION & STORAGE ENGINE | | 1. MIME & Size Validation

(.jpg, .png, .mp4, .webm, .mov) | | 2. Content-Addressable Storage Write -> ./uploads/{hash}.ext |

| 3. Register Media Record in animal\_health.db (Type: 'video' | 'image') |

+-----

+-----+ | POST /api/health-check/analyze { media\_id:

11, symptoms: [...] } v

+-----

-----+ | TEMPORAL KEYFRAME EXTRACTION (FFmpeg) | | If video -> ffmpeg -y -i input.webm -vf

"fps=1/2,scale=512:-1" -vframes 4 frame\_%03d.jpg | | Collect 4 base64 data URLs:

["data:image/jpeg;base64,...", "data:image/jpeg;base64,..."] |

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+-----+ | Payload: 4 Image Data URLs + Unbiased

| Objective                                                                                                                                                                                                                                                                                       | Inspection | Prompt | v       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|---------|
| +-----+   STAGE 1: MULTIMODAL COMPUTER VISION AI     Model: Qwen/Qwen2.5-VL-72B-Instruct (Vision)                                                                                                                                                                                               |            |        |         |
| Outputs JSON: species, posture, visible lesions, discharges, wounds, gait/chorea, behavior, severity                                                                                                                                                                                            |            |        |         |
| +-----+   Payload: Species + Vision Findings +                                                                                                                                                                                                                                                  |            |        |         |
| Clinical                                                                                                                                                                                                                                                                                        | Complaints | /      | History |
| +-----+   STAGE 2: VETERINARY INTERNAL MEDICINE REASONING     Model: Qwen/Qwen2.5-72B-Instruct (Clinical Specialist)     Outputs JSON: 4-5 ranked differentials, pathophysiological reasons, confirmatory tests, triage urgency                                                                 |            |        |         |
| +-----+   Persistence & Automated Backup Event Hook (OnChange)                                                                                                                                                                                                                                  |            |        |         |
| +-----+   DATABASE PERSISTENCE & ASYNC BACKUP ENGINE     1. Auto-resolve & persist Animal Record (linked to authenticated User)     2. Insert Health Screening Record in animal_health.db     3. Trigger OnChange() -> Asynchronous Snapshot Backup via VACUUM INTO -> ./backups/{timestamp}.db |            |        |         |
| +-----+   JSON Response (HTTP 201 Created) -> Render                                                                                                                                                                                                                                            |            |        |         |

## 6. Detailed In-Depth Pipeline Mechanics

### 6.1. Temporal Video Keyframe Extraction & Quota Compliance

Modern Multimodal LLM endpoints (e.g. Hugging Face Router for `Qwen2.5-VL-72B-Instruct`) enforce strict limits of **at most 5 images per request prompt**. Videos recorded in browsers or uploaded from phones often contain hundreds of frames at 30–60 FPS.

The backend implements adaptive temporal sampling in `backend/internal/ai/providers.go`:

```
// Extract 4 representative temporal keyframes across the video using ffmpeg
// Enforces max 5 image quota constraint on the multimodal inference router

cmd := exec.CommandContext(ctx, "ffmpeg", "-y", "-i", cleanPath,
    "-vf", "fps=1/2,scale=512:-1", "-vframes", "4", framePattern)

if err := cmd.Run(); err == nil {

    files, _ := filepath.Glob(filepath.Join(tmpDir, "frame*.jpg"))

    sort.Strings(files)

    for _, f := range files {

        data, _ := os.ReadFile(f)

        frames = append(frames, "data:image/jpeg;base64," +
base64.StdEncoding.EncodeToString(data))

        if len(frames) ≥ 4 { break }

    }

}
```

## 6.2. Unbiased Dual-Stage Clinical AI Reasoning

To ensure the AI evaluates cases on a true medical spectrum without artificial disease hallucinations or forced diagnoses:

- **Stage 1 (Vision):** Strictly extracts physical abnormalities (e.g. `"lateral recumbency, bandaged right front paw, fluid/hematochezia stain near perineum, no voluntary movement"`). No disease names are injected into the prompt.
- **Stage 2 (Veterinary Reasoning):** Evaluates evidence across infectious (viral, bacterial, parasitic), mechanical (foreign bodies, obstructions), toxicological, inflammatory, and metabolic categories.

### 💡 Unbiased Diagnostic Guidelines (System Prompt)

"1. UNBIASED EVALUATION: Formulate differentials based purely on patient species, reported complaints, and visual findings.  
2. COMPREHENSIVE SPECTRUM: Systematically consider all relevant etiologies across infectious (viral, bacterial, fungal, parasitic), mechanical/surgical (obstructions, trauma), toxicological, inflammatory, and metabolic disorders.  
3. EVIDENCE-BASED RANKING: Provide 4-5 differential diagnoses ranked by true clinical likelihood with pathophysiological reasoning.  
4. DIAGNOSTIC WORKUP & TRIAGE: Recommend specific confirmatory lab tests, imaging modalities, and immediate emergency management."

## 6.3. Concurrency Resilience & Lock-Free SQLite WAL Setup

To allow concurrent read operations, write operations, and background snapshot auto-backups without `database is locked` exceptions, the SQLite connections are configured with explicit busy timeouts and single-writer pooling:

```
// Connection string with busy timeout and WAL journal mode
db, err := sql.Open("sqlite3", path + "?_foreign_keys=on&_journal_mode=WAL&_busy_timeout=10000")
db.SetMaxOpenConns(1)
db.Exec(`PRAGMA foreign_keys = ON; PRAGMA journal_mode = WAL; PRAGMA busy_timeout = 10000;`)
```

## 7. Database Schemas & Migration Architecture

The platform manages two segregated SQLite databases with automated versioned migrations applied

on every server startup:

| Database                                      | Table Name        | Columns & Structural Schema                                                                                                                  |
|-----------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| user.db<br>Authentication & Security          | users             | id (PK), email (UNIQUE), password_hash (Bcrypt), full_name, role, created_at, updated_at                                                     |
|                                               | schema_migrations | version (PK), name, applied_at                                                                                                               |
|                                               | animals           | id (PK), user_id (INDEX), name, species, breed, age, sex, photo_url, notes, weight, created_at                                               |
|                                               | media             | id (PK), url, path, type ('image' 'video'), mime_type, size, created_at                                                                      |
| animal_health.db<br>Clinical Application Data | health_screenings | id (PK), animal_id (INDEX), media_url, media_type, symptoms_json, visual_analysis_json, assessment_json, urgency (INDEX), created_at (INDEX) |
|                                               | reminders         | id (PK), user_id (INDEX), animal_id, type, title, description, due_at, completed, created_at                                                 |
|                                               | schema_migrations | version (PK), name, applied_at                                                                                                               |

## 8. RESTful API Endpoint Specifications

| Method & Path           | Headers & Parameters           | Description & Payload Structure                                                                                                            |
|-------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| POST /api/auth/register | Content-Type: application/json | Registers a new user account with Bcrypt password hashing (Cost 10).<br>Payload: { "email": "...", "password": "...", "full_name": "..." } |
| POST /api/auth/login    | Content-Type: application/json | Authenticates credentials against user.db and issues session token.<br>Response: { "token": "...", "user": { "id": 1, "email": "..." } }   |
| POST /api/auth/logout   | X-Auth-Token: <token>          | Invalidates active user session token.                                                                                                     |

| Method & Path                  | Headers & Parameters                                         | Description & Payload Structure                                                                                                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST /api/health-check/upload  | Content-Type: multipart/form-data<br>X-Auth-Token, X-User-ID | Uploads image or video file. Validates MIME type, stores file in ./uploads/, and registers record in media table.<br>Response: { "id": 11, "url": "/uploads/...", "type": "video", "size": 1938155 }                                                                         |
| POST /api/health-check/analyze | Content-Type: application/json<br>X-Auth-Token, X-User-ID    | Executes full multimodal visual inspection and veterinary differential assessment.<br>Payload: { "media_id": 11, "symptoms": { "symptoms": [...], "other": "..." } }<br>Response: Full structured JSON with visual analysis, assessment, differential rankings, and urgency. |
| GET /api/health-screenings/:id | X-Auth-Token                                                 | Retrieves complete persisted diagnostic report by ID.                                                                                                                                                                                                                        |
| GET /api/animals               | X-Auth-Token                                                 | Lists all registered patients and profiles for the authenticated user.                                                                                                                                                                                                       |
| POST /api/animals              | Content-Type: application/json<br>X-Auth-Token               | Creates a new patient animal profile.                                                                                                                                                                                                                                        |
| GET /api/reminders             | X-Auth-Token                                                 | Lists active vaccination, medication, and clinical follow-up reminders.                                                                                                                                                                                                      |

## 9. Security Architecture & Reliability Engineering

### Security & Zero-Leakage Architecture

- **Header Isolation:** User session tokens are strictly passed over X-Auth-Token and X-User-ID headers. The server's AI client injects the upstream Hugging Face API key independently via Authorization: Bearer <HUGGINGFACE\_API\_KEY>, preventing session token leaks and 401 collisions.
- **No Public Admin Endpoints:** All database inspection, health checks (PRAGMA integrity\_check), and automated backup triggers are strictly internal to the server process and are never exposed to the public Internet.
- **Bcrypt Password Security:** User passwords in user.db are salted and hashed using adaptive Bcrypt (cost factor 10). Plaintext passwords are never stored in memory or on disk.

- **Atomic Point-in-Time Backups:** The backup engine uses native transactional `VACUUM INTO` commands, producing zero-downtime, fully verified, uncorrupted backup snapshots in `./backups/` upon every database mutation.

## 10. Verified Clinical Case Studies & Test Benchmarks

| Test Input & Scenario                                                                                     | AI Vision Detection                                                                                              | AI Differential Diagnoses                                       | Triage Urgency |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------|
| <b>Puppy</b><br><b>Video</b><br>WebM video clip<br>barking at kitchen bin                                 | <b>Playful</b><br>Alert, playful posture, voluntary movements, no skin/mucosal lesions, no respiratory distress. | 1. <b>[HIGH]</b> Normal Puppy Play Behavior                     | LOW            |
|                                                                                                           |                                                                                                                  | 2. <b>[POSSIBLE]</b> Behavioral Response to Reflection          |                |
|                                                                                                           |                                                                                                                  | 3. <b>[LOW]</b> Attention-seeking behavior                      |                |
| <b>Canine</b><br><b>Neurological Motion</b><br>Involuntary twitching, hyperkeratosis, crusting, hard pads | / Rhythmic jaw/limb myoclonus, ocular-nasal discharge, paw pad                                                   | 1. <b>[HIGH]</b> Canine Distemper Virus (CDV)                   | HIGH           |
|                                                                                                           |                                                                                                                  | 2. <b>[MODERATE]</b> Immune-Mediated Polyneuropathy             |                |
|                                                                                                           |                                                                                                                  | 3. <b>[POSSIBLE]</b> Zinc-Responsive Dermatitis                 |                |
| <b>Adult Dog with Trauma</b><br>dog.jpg (No symptoms supplied)                                            | Lateral recumbency, bandaged right front paw, perineal fluid stain on floor.                                     | 1. <b>[HIGH]</b> Traumatic Paw Injury / Post-Op                 | / MODERATE     |
|                                                                                                           |                                                                                                                  | 2. <b>[MODERATE]</b> Urinary Tract Infection                    |                |
|                                                                                                           |                                                                                                                  | 3. <b>[POSSIBLE]</b> Perineal Hernia / Drainage                 |                |
| <b>Critical Hemorrhagic Presentation</b><br>dog.jpg + 24h bloody diarrhea vomiting                        | Lateral recumbency, severe prostration, continuous hematochezia fluid on floor, & bandaged leg.                  | 1. <b>[HIGH]</b> Canine Parvovirus (CPV-2)                      | EMERGENCY      |
|                                                                                                           |                                                                                                                  | 2. <b>[MODERATE]</b> Acute Hemorrhagic Diarrhea Syndrome (AHDS) |                |
|                                                                                                           |                                                                                                                  | 3. <b>[POSSIBLE]</b> Intestinal Foreign Body Obstruction        |                |

WAL & Multimodal AI